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RESEARCH ARTICLE

Skin Lesion Segmentation in Clinical Images Using Nonnegative Matrix Factorization and Mutual Information

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ABSTRACT Early detection of cutaneous melanoma is crucial for improving patient survival rates. Skin lesion segmentation is a critical step towards computer-aided diagnosis that is effective in melanoma detection. This work proposes a novel patch-based model that leverages a combination of non-negative matrix factorization and mutual information for segmenting cutaneous melanomas in clinical images. The proposed approach tends to lead to accurate skin lesion segmentations, and it was comprehensively evaluated in terms of its effectiveness based on three benchmark datasets (DermIS, DermNet, and DermQuest) using statistical analysis, ablation studies, and cross-dataset testing. The proposed method tends to improve on the segmentation results provided by established deep learning baselines (FCNet, FCN, PSPNet, SegNet, U-Net) and by recent advancements specifically designed for skin lesion segmentation (DSNet, FrCNet, and LGPNet).

INDEX TERMS Biomedical image processing, image segmentation, matrix decomposition, skin cancer.

I. INTRODUCTION

Cutaneous melanoma, a malignant neoplasm arising from melanocytes, can be a lethal form of skin cancer due to its proclivity for metastasis via the lymphatic and blood systems. According to data from the Global Cancer Observatory (GLOBOCAN) and the International Agency for Research on Cancer (IARC), the number of cutaneous melanoma cases is projected to increase from 325,000 to 510,000, with the associated mortality rate rising from 57,000 to 96,000, between 2020 and 2040 [1].

In this context, early diagnosis of cutaneous melanoma is crucial for improving patient prognosis. Specifically, when the disease remains localized, the survival rate is near 100%. However, this rate drops considerably if melanoma reaches the lymph nodes, and plummets to around 35% when distant organs are involved [2]. In practice, dermatologists use a non-invasive imaging technique known as dermoscopy to distinguish melanomas from visually similar benign skin lesions (e.g., atypical nevi) [3]. Essentially, dermoscopy relies on the use of a digital dermatoscope, which is a magnification tool often equipped with polarized illumination, to reveal discriminative subsurface structures such as pigment networks, dots and globules. Prior to analysis, a contact medium (i.e., a fluid), such as a gel or immersion oil, is commonly applied to the lesion site in order to mitigate the light reflection effects [4].

Despite its advantages for timely diagnosis, dermoscopy is not commonplace in primary healthcare, particularly in remote areas and developing countries, where appointments with dermatologists frequently require long waiting times. To improve this scenario, several computer-aided diagnosis (CAD) systems have been proposed for dermoscopic images [5]. Nevertheless, such approaches may be of limited applicability when only lower-quality images (e.g., captured by smartphones) are available, which is a more likely to be the case in resource-limited settings [6].

Alternative CAD systems designed to work with clinical (or macroscopic) images offer a broader spectrum of applications, encompassing pre-screening patients in primary

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healthcare followed by consultations with a general practitioner [7], [8]. Nonetheless, the effectiveness of these systems depends critically on automatic image segmentation [9], [10], [11], [12], [13], [14]. It shall be observed that automatic image segmentation techniques have also been applied successfully to other medical imaging modalities (e.g., see [15] and/or [16]).

Automatic image segmentation involves identifying the skin lesion of interest within a clinical image by precisely differentiating the pixels belonging to the lesion from those belonging to the background, which may include healthy skin, hair, and artifacts introduced during the image acquisition process. Notably, segmentation is a crucial component in the aforementioned systems, enabling the extraction of clinically relevant features, such as lesion asymmetries, boundaries, color variations, and diameter, while preventing mistaken classification based on spurious (non-causal) data [17].

In the last decades, both traditional similarity-based (e.g., thresholding, clustering, region growing) and discontinuity-based approaches (e.g., edge detection, watershed, active contours) have played a key role in developing advanced techniques for skin lesion segmentation. In spite of these advancements, segmenting skin lesions still is an open problem due to both, intra-class variability (e.g., variations in skin tones across patients and colors/textures in lesions), and inter-class similarities (e.g., lesions resembling healthy skin), as comprehensively reviewed in [18].

Clinical images, in comparison with dermoscopic images, exhibit additional segmentation challenges, such as reduced spatial resolution, uneven illumination leading to shadow and reflection effects, and increased image artifacts. Fig. 1 exemplifies some of these challenges, displaying three skin lesions captured using both imaging modalities. The top row shows the lesions as they appear in clinical images Figs. 1a, 1c and 1e), while the bottom row presents the lesions as they appear in dermoscopic images Figs. 1b, 1d and 1f).

Recently, advancements in deep learning enabled the emergence of a new class of approaches, which tailor deep models (e.g., a convolutional neural network – CNN) for segmentation tasks. These approaches leverage architectures like Fully Convolutional Network (FCNet), U-shaped Net (U-Net) [20], Pyramid Scene Parsing Network (PSPNet) [21], and Segmentation Net (SegNet) [22]. An updated survey detailing these architectures and their specialized variants in the context of skin lesion segmentation can be found in [23].

This work presents a novel clustering-based method for precise skin lesion segmentation in clinical images, and its key contributions are the following:

- A new patch-based model that combines nonnegative matrix factorization (NMF) [24] with mutual information [25] is presented;
- A scheme is proposed to efficiently integrate the normalized graph cuts (NGC) [26] technique with the proposed patch-based model to enhance segmentation accuracy and efficiency;
- An extensive experimental evaluation of the proposed approach is conducted on the DermIS [27], DermNet [28], and DermQuest [27] datasets, involving statistical analyses, ablation studies, and cross-dataset tests;
- Based on the experimental results, the proposed scheme tends to provide an improved performance compared to both, deep learning baselines (FCNet, FCN, PSPNet, SegNet, U-Net) and more recent advancements specifically designed for skin lesion segmentation (e.g., DSNet [29], FrCNet [30], and LGPNet [31]).

The remainder of this paper is structured as follows. Section II details the proposed skin lesion segmentation method. Section III then discusses the experimental results obtained with the proposed method. Finally, Section IV presents the conclusions drawn from this study.

II. PROPOSED APPROACH FOR SKIN LESION SEGMENTATION USING NONNEGATIVE MATRIX FACTORIZATION AND MUTUAL INFORMATION

The proposed skin lesion segmentation method provides a binary mask indicating the location of the skin lesion regions for a given input clinical image. The workflow of the proposed scheme for skin lesion segmentation is illustrated in Fig. 2.

Next, the steps of the proposed skin lesion segmentation method are detailed in Section II-A, to II-D, followed by an analysis of its computational complexity in Section II-E.

A. EXTRACTING COLOR SALIENCIES FROM THE INPUT SKIN LESION IMAGE

The input image is first pre-processed to mitigate uneven illumination effects using the *shading attenuation* method, as described in [32]. Subsequently, a color saliency map is obtained using the method detailed in [33], which accentuates the chromatic differences between healthy and unhealthy skin pixels. Examples of such images are provided in Fig. 3.

B. DICTIONARY LEARNING VIA NONNEGATIVE MATRIX FACTORIZATION AND MUTUAL INFORMATION

The color saliency map, $\mathcal{S} \in \mathbb{R}^{H \times W}$, is partitioned into N non-overlapping patches, each with dimensions $\omega \times \omega$. These patches can be properly arranged in a patch matrix, $\mathcal{P} = [\mathbf{p}_1, \ldots, \mathbf{p}_N]$, where each $\mathbf{p}_j \in \mathbb{R}^M$ ($M = \omega^2$), representing a one-dimensional vector obtained by stacking the elements of the corresponding patch in a column-wise manner, is indexed by $j \in \{1, \ldots, N\}$. The primary hyperparameter of the proposed method is K, as detailed below, and for a given $K < \min\{M, N\}$ the patches matrix $\mathcal{P}$ is further decomposed into two matrices by applying the alternating least squares (ALS) method for nonnegative matrix factorization (NMF) [34]:

$$(\mathcal{D}^0, \mathcal{X}^0) = \arg\min_{\mathcal{D},\mathcal{X}} \left\{ \frac{1}{2} \|\mathcal{P} - \mathcal{D}\mathcal{X}\|_F^2 \right\}. \quad (1)$$

In Equation (1), $\mathcal{D}^0 = [\mathbf{d}_1^0, \ldots, \mathbf{d}_K^0] \in \mathbb{R}^{M \times K}$ can be seen as an initial dictionary, where each column $\mathbf{d}_k^0 \in \mathbb{R}^M$, referred

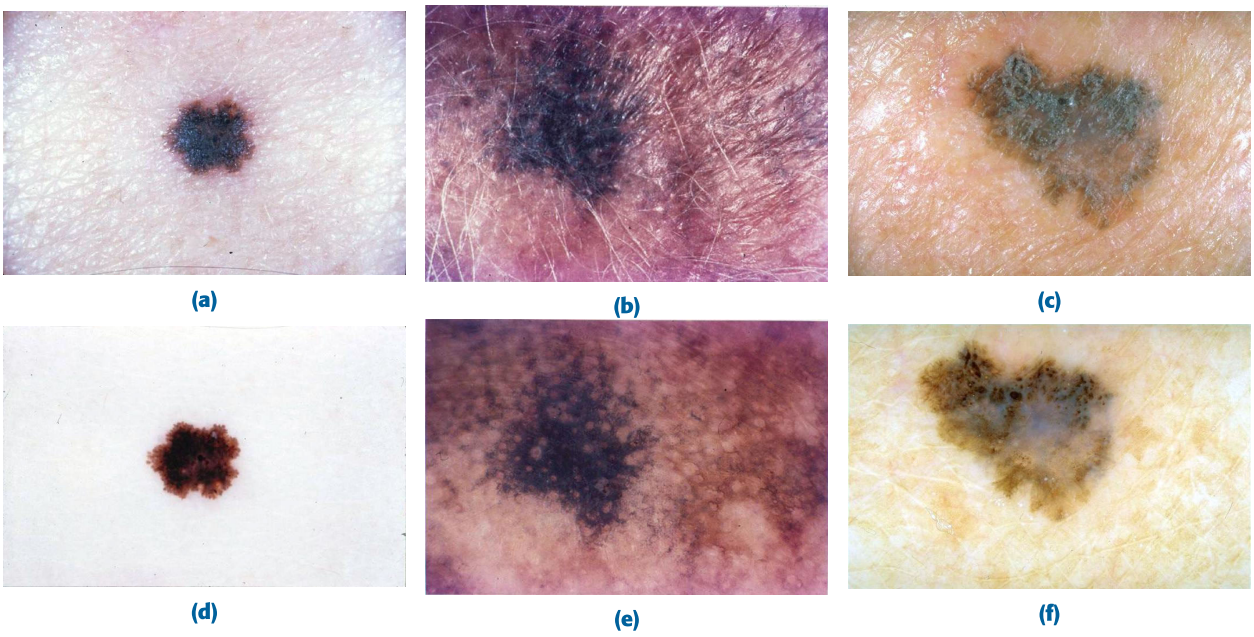


FIGURE 1. Comparison between dermoscopic and macroscopic images captured from the same pigmented skin lesions [19]. The top row (i.e., Fig. 1a, Fig. 1b, and Fig. 1c) displays macroscopic images of pigmented skin lesions, which also are exhibited with the corresponding dermoscopic images in the bottom row (i.e., Fig. 1d, Fig. 1e, and Fig. 1f).

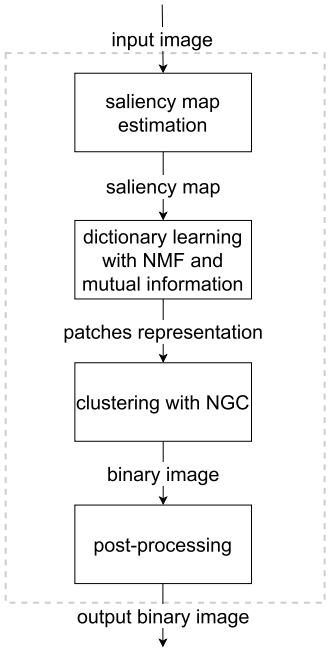


FIGURE 2. Overview of the proposed skin lesion segmentation method.

to as *an atom* of $\mathcal{D}^0$, is indexed by $k \in \{1, \ldots, K\}$, with K indicating the number of *dictionary atoms*. Consequently, each patch in $\mathcal{P}$ is approximated by a linear combination of the atoms in $\mathcal{D}^0$. Therefore, K specifies the number of *atoms* $\mathbf{d}_k^0$ of the initial dictionary $\mathcal{D}^0$ that are going to be used for approximating the patches in $\mathcal{P}$. Moreover, $\mathcal{X}^0 = [\mathbf{x}_1^0, \ldots, \mathbf{x}_N^0] \in \mathbb{R}^{K \times N}$, indicates the weights vectors that are going to be assigned to the K*atoms* $\mathbf{d}_k^0$ in the patches approximations. Recall that each *atom* $\mathbf{d}_k^0$ and patch $\mathbf{p}_j$ is M-dimensional ($\mathbf{d}_k^0$ and $\mathbf{p}_j \in \mathbb{R}^M$). Thus, the j-th column of $\mathcal{X}^0$ is $\mathbf{x}_j^0 = (x_{1,j}, \ldots, x_{K,j}) \in \mathbb{R}^K$ and contains the K weights of the K atoms in approximating $\mathbf{p}_j$ such that:

$$\mathbf{p}_j \simeq \sum_{k=1}^{K} x_{k,j} \mathbf{d}_k^0, \text{ for } j = 1, \ldots, N. \tag{2}$$

Although $\mathbf{x}_j^0$ can serve as a feature vector to represent $\mathbf{p}_j$ [35], the method proposed herein makes use of a lower-dimensional vector. In other words, instead of representing the image patches using $\mathcal{D}^0$ directly, the proposed method utilizes a more compact and representative dictionary, which consists of a subset of K atoms of $\mathcal{D}^0$.

In this work, mutual information is employed to model two relevant dictionary features, namely, its compactness and representativeness. Starting with the initial dictionary $\mathcal{D}^0$, an adequate subset of K *atoms* is selected to compose the refined dictionary $\mathcal{D}^*$ based on the following optimization problem:

$$\mathcal{D}^* = \underset{\mathcal{D} \subseteq \mathcal{D}^0}{\arg\max} \left\{ \lambda_1 \mathbb{I}(\mathcal{D}, \mathcal{D}^0 - \mathcal{D}) + \lambda_2 \mathbb{I}(\mathcal{D}, \mathcal{P}) \right\}, \tag{3}$$

where, $\mathbb{I}(\mathcal{D}, \mathcal{D}^0 - \mathcal{D})$ represents the mutual information between $\mathcal{D}$ and $\mathcal{D}^0$–$\mathcal{D}$ (i.e., the compactness of the candidate dictionary $\mathcal{D}$), which quantifies the information encoded by the *atoms* selected to form $\mathcal{D}$ with respect to the other *atoms* remaining in $\mathcal{D}^0$, that were not chosen to compose $\mathcal{D}$. Also, $\mathbb{I}(\mathcal{D}, \mathcal{P})$ is the mutual information between $\mathcal{D}$ and $\mathcal{P}$ (i.e., the representativeness of $\mathcal{D}$), and it measures the extent to which

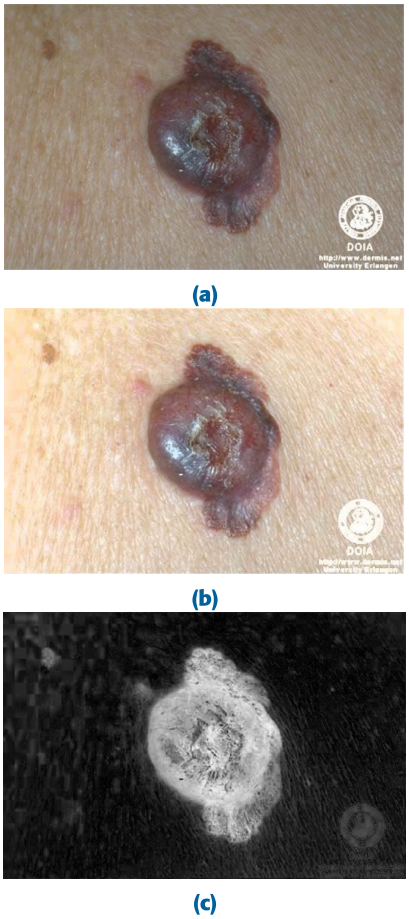


FIGURE 3. Pre-processing results for an illustrative pigmented skin lesion image. (a) Original image. (b) Illumination-corrected image. (c) Color saliency map.

$\mathcal{D}$ captures information about the overall set of patches in $\mathcal{P}$. Furthermore, λ_1 and λ_2 are hyperparameters that balance the importance of the compactness and representativeness features in the objective function in Equation (3).

It shall be observed that the problem described in Equation (3) is NP-hard, but it can be efficiently approximated in a greedy near-optimal fashion [36]. However, to avoid introducing an additional hyperparameter with the use of a fixed dictionary size, in this work it is proposed to employ a variant of the method introduced in [36]. This variant introduces an adaptive dictionary selection strategy which adjusts dynamically the number of *dictionary atoms*, and can be summarized as follows:

1) Initialize $\tilde{\mathcal{D}}$ as an empty set ($\tilde{\mathcal{D}} = \varnothing$);
2) In each iteration $t \in \{1, 2, \ldots, K\}$, choose an *atom* from $\mathcal{D}^0 - \tilde{\mathcal{D}}$ to be the t-th column of $\tilde{\mathcal{D}}$ according to:

$$\mathbf{d}_t^* = \arg\max_{\mathbf{d}_t \in \mathcal{D}^0 - \tilde{\mathcal{D}}} \left\{ \lambda_1 \left[\mathbb{I}\left(\tilde{\mathcal{D}} \cup \mathbf{d}_t, \mathcal{D}^0 - (\tilde{\mathcal{D}} \cup \mathbf{d}_t)\right) - \mathbb{I}\left(\tilde{\mathcal{D}}, \mathcal{D}^0 - \tilde{\mathcal{D}}\right) \right] + \lambda_2 \left[\mathbb{I}\left((\tilde{\mathcal{D}} \cup \mathbf{d}_t), \mathcal{P}\right) - \mathbb{I}\left(\tilde{\mathcal{D}}, \mathcal{P}\right) \right] \right\}. \tag{4}$$

3) Determine the last iteration t for which the maximum value of the objective function in Equation (4) is positive, and this last iteration is denoted as t^*;
4) Build the optimized dictionary, $\tilde{\mathcal{D}}_{t^*}$, which contains the first t^* columns of $\tilde{\mathcal{D}}$;
5) Obtain the representations of the patches in $\mathcal{P}$ using $\tilde{\mathcal{D}}_{t^*}$:

$$\mathbf{X} = \tilde{\mathcal{D}}_{t^*}^{\dagger} \mathcal{P}, \tag{5}$$

where $\tilde{\mathcal{D}}_{t^*}^{\dagger}$ is the pseudo-inverse of $\tilde{\mathcal{D}}_{t^*}$.

To estimate the mutual information quantities involved in Equation (4), all sets provided as input to the correspondent problem, and consequently, their subsets, are modeled by Gaussian stochastic processes (i.e., the elements of these sets are associated with random variables with Gaussian distributions) [37]. Additionally, it is noteworthy that the parameters λ_1 and λ_2 can be directly estimated in a data-driven manner. For further details, please refer to [36].

C. SEGMENTING SKIN LESIONS BY COMBINING DICTIONARY-BASED PATCH REPRESENTATION WITH NORMALIZED GRAPH CUTS

The representations of the patches in $\mathcal{X}^*$ can be regarded as vertices of a weighted undirected graph, where the edge weights encode the pairwise similarity between patches measured by:

$$\mathcal{E}(\mathbf{x}_j^*, \mathbf{x}_{j'}^*) = \exp\left(-\frac{1}{2\sigma_e^2} \|\mathbf{x}_j^* - \mathbf{x}_{j'}^*\|_2^2\right), \tag{6}$$

where σ_e controls the width of the Gaussian kernel.

Using these edge weights, the normalized graph cuts method [26] is applied to segment the image by partitioning the image patches into two distinct clusters: $\mathcal{A}^*$ and $\mathcal{B}^* = \mathcal{X}^* - \mathcal{A}^*$. Specifically, the image segmentation by patch clustering is solving the following function minimization problem:

$$(\mathcal{A}^*, \mathcal{B}^*) = \arg\min_{\mathcal{A},\mathcal{B}} \left\{ \frac{\text{cut}(\mathcal{A}, \mathcal{B})}{\text{assoc}(\mathcal{A}, \mathcal{X}^*)} + \frac{\text{cut}(\mathcal{A}, \mathcal{B})}{\text{assoc}(\mathcal{B}, \mathcal{X}^*)} \right\}, \tag{7}$$

where:

$$\text{cut}(\mathcal{A}, \mathcal{B}) = \sum_{(\mathbf{a},\mathbf{b}) \in \mathcal{A} \times \mathcal{B}} \mathcal{E}(\mathbf{a}, \mathbf{b}), \tag{8}$$

$$\text{assoc}(\mathcal{A}, \mathcal{X}^*) = \sum_{(\mathbf{a},\mathbf{x}^*) \in \mathcal{A} \times \mathcal{X}^*} \mathcal{E}(\mathbf{a}, \mathbf{x}^*), \tag{9}$$

and

$$\text{assoc}(\mathcal{B}, \mathcal{X}^*) = \sum_{(\mathbf{b},\mathbf{x}^*) \in \mathcal{B} \times \mathcal{X}^*} \mathcal{E}(\mathbf{b}, \mathbf{x}^*). \tag{10}$$

Afterwards, the binary segmentation mask $\mathcal{M}^0 \in \{0, 1\}^{H \times W}$ is obtained, such that $\mathcal{M}^0(r, c) = 0$ if the pixel indexed by (r, c) belongs to a patch clustered in the partition $\mathcal{A}^*$, or $\mathcal{M}^0(r, c) = 1$ if the pixel indexed by (r, c) belongs to a patch clustered in the partition $\mathcal{B}^*$. It shall be observed that at

this stage, the image pixels have the same label as the patch they belong to.

D. POST-PROCESSING

In order to label the image pixels more consistently, the color saliency map is used to obtain a standardized mask $\mathcal{M} \in \{0,1\}^{H \times W}$ as follows:

$$\mathcal{M}(r,c) = \begin{cases} 1 - \mathcal{M}^0(r,c), & \text{if } \sum_{(r',c') \in \mathbb{V}} \mathcal{S}(r',c') > \frac{1}{2}|\mathbb{V}|, \\ \mathcal{M}^0(r,c), & \text{if } \sum_{(r',c') \in \mathbb{V}} \mathcal{S}(r',c') \leq \frac{1}{2}|\mathbb{V}|, \end{cases} \tag{11}$$

with

$$\begin{aligned} \mathbb{V} = \{(r,c) \in \mathrm{dom}(\mathcal{S}) : & \ 1 \leq r \leq \kappa \text{ and } 1 \leq c \leq \kappa \text{ or} \\ & 1 \leq r \leq \kappa \text{ and } W - \kappa + 1 \leq c \leq W \text{ or} \\ & H - \kappa + 1 \leq r \leq H \text{ and } 1 \leq c \leq \kappa \text{ or} \\ & H - \kappa + 1 \leq r \leq H \text{ and } W - \kappa + 1 \leq c \leq W\} \end{aligned} \tag{12}$$

denoting the set of ordered pairs within the four corner regions of $\mathcal{S}$, and κ is the square area covered by these corner regions.

Since $\mathcal{M}$ may comprise multiple connected components corresponding to distinct skin lesions, only the *primary component* (or *region of interest*) is selected to represent the lesion of interest, with its location determined by:

$$\mathbb{L}_u^* = \arg\max_{\mathbb{L}_u} \left\{ \sum_{(r,c) \in \mathbb{L}_u} \exp\left(-\frac{(r - r_0)^2 + (c - c_0)^2}{2\sigma_l^2} \right) \right\}, \tag{13}$$

where $\mathbb{L}_u \in \{\mathbb{L}_h\}_{u=1}^{u=L}$ denotes a set of ordered pairs containing the coordinates of all pixels within a particular connected component. Here, the objective function increases as the connected component expands and/or approaches the image center, $(r_0, c_0) = (\lfloor H/2 \rfloor, \lfloor M/2 \rfloor)$, and σ_l determines the spread of the Gaussian kernel.

Finally, additional post-processing steps are applied to smooth the lesion contours, while reducing the ''blocky'' contours appearance that resulted from using non-overlapping square patches. In particular, the detected boundaries are smoothed via Gaussian filtering, followed by global thresholding with a fixed threshold value. Subsequently, if holes are present within the segmented skin lesion region, they are morphologically closed using a flood fill algorithm [38].

Fig. 4 provides a step-by-step visual illustration of the post-processing pipeline for a pigmented skin lesion image.

E. COMPUTATIONAL COMPLEXITY

In the proposed approach, the input image is divided into square patches with their side lengths automatically defined as the odd integer closest to 1% of the larger image dimension, according to the following equation:

$$\omega = \left\lfloor \frac{0.01L + 2}{2} \right\rfloor - 1, \tag{14}$$

where $L = \max\{H, W\}$, with W and H denoting, respectively, the number of rows and columns of the input image. Thus, the number of pixels per patch, M, and the total number of patches, N, can be expressed as functions of L.

Considering $L \in [300, 5000)$, which encompasses typical sizes of dermatological images, the plots in Fig. 5 show how M and N vary as a function of L. As shown in these plots, for typical values of L, $M \leq L/2 = \mathcal{O}(L)$ and $N \leq 18,000 = \mathcal{O}(1)$. Therefore, each stage of the proposed pipeline has a computational complexity that is summarized in Table 1.

In light of the previous discussion, the proposed method objectively achieves an overall computational complexity of $\mathcal{O}(L^2)$, although the implicit constants may not be negligible (e.g., can be of the order of $10000^3 = 10^8$ in the normalized graph cut stage).

III. EXPERIMENTAL RESULTS

To evaluate the effectiveness of the proposed method, experiments were conducted on three datasets: DermIS [27], containing 69 macroscopic images (43 melanoma cases and 26 atypical nevus cases); DermNet [28], containing 152 macroscopic images (107 melanoma cases and 45 atypical nevus cases); and DermQuest [27], containing 137 macroscopic images (76 melanoma cases and 61 atypical nevus cases).

The comparisons among methods involved the evaluation for multiple pixel sets, and were performed using the Friedman test with the Nemenyi post-hoc test for all pairwise method comparisons. Also, the Bonferroni-Dunn post-hoc test was conducted when comparing a control method (e.g., a proposed method) against the alternatives [39]. A confidence level of 95% was selected in the experiments.

A. EXPERIMENTAL SETUP

The measure chosen for a comprehensive assessment of skin lesion segmentation performances of the comparative methods is the BF_1 score [40]. This contour-based measure is deemed more convenient than the region-based measures traditionally employed [41], which treat segmentation as a pixel-level classification and do not produce similar results even for two segmentation outcomes that differ only in the labeling of the lesion pixels as ones or zeros.

It shall be clarified that the proposed method has only one tunable hyperparameter, namely K, and all other parameters are fixed or adaptively derived from the input image, as summarized in Table 2. Moreover, all experiments were conducted on a laptop without a dedicated GPU, equipped with an Intel i3-4030U CPU (1.90 GHz) and 12 GB of RAM.[1]

1) CROSS-DATASET EXPERIMENTAL DESIGN

To mitigate the system vulnerability to dataset biases, the proposed segmentation method does not rely on explicit

[1] All codes and additional details necessary to replicate the experiments presented herein are publicly available at: https://github.com/eliezersflores/skinseg-nmfmi

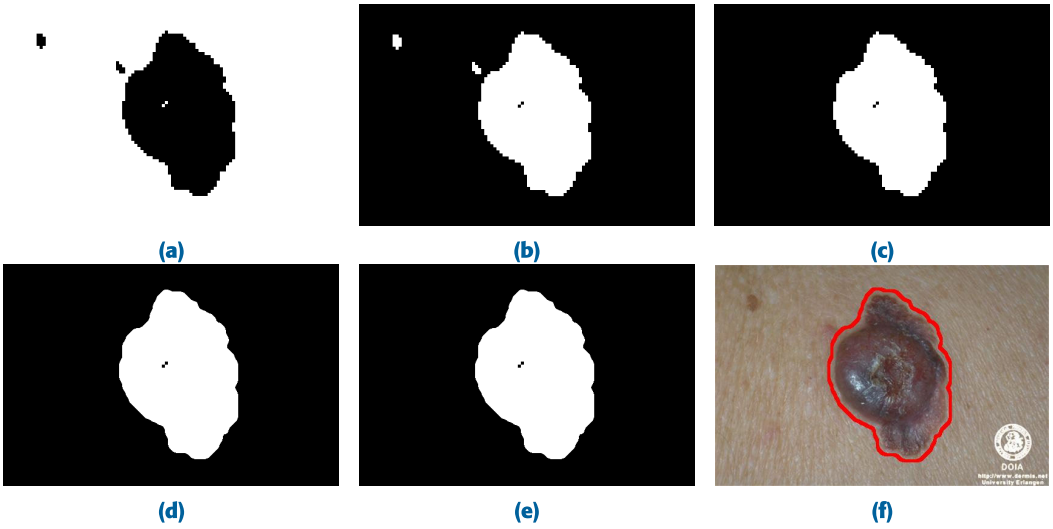
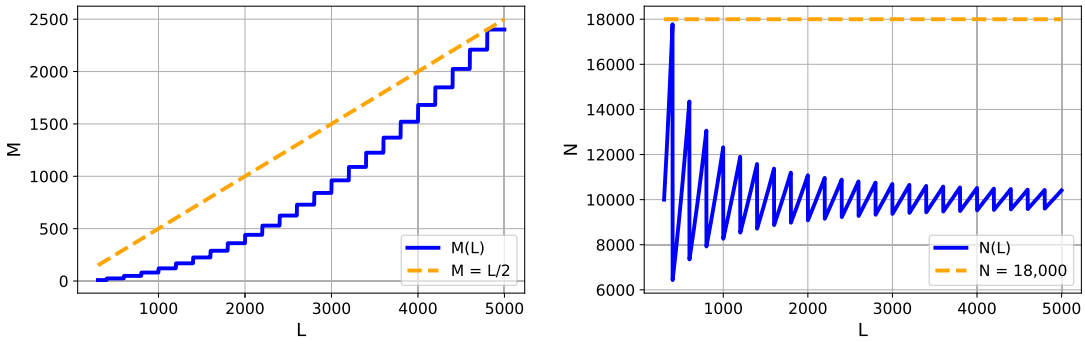


FIGURE 4. Illustration of the segmentation and post-processing results obtained for a pigmented skin lesion image. (a) Initial binary segmentation mask. (b) Standardized mask. (c) Selected lesion of interest. (d) Lesion after boundary smoothing. (e) Lesion post-flood fill. (f) Final segmentation result overlaid on the original image.



(a) M **as a function of** L. **(b)** N **as a function of** L.

FIGURE 5. Estimates of the number of pixels per patch (M) and of the total number of patches (N) by changing the maximum image dimension (L).

TABLE 1. Computational complexity of the main steps of the proposed skin lesion segmentation pipeline, expressed as functions of the larger image dimension L.

Stage	Computational complexity
Saliency map estimation	$\mathcal{O}(L^2)$
Dictionary learning with NMF and mutual information	$\mathcal{O}(L)$
Clustering with NGC	$\mathcal{O}(1)$
Post-processing	$\mathcal{O}(L^2)$

training, but instead it uses unsupervised learning for lesion pixel identification.

Therefore, in Section III-B is presented how the critical hyperparameter K was initially set, and then fine-tuned through a straightforward grid search on the DermIS dataset, which is the smallest database among those considered in this study.[2] Subsequently, as elaborated in Section III-C, a performance comparison between the proposed method and its ablated versions on the DermIS dataset is presented. Finally, as described in Section III-D, a performance comparison of state-of-the-art alternatives with the proposed method was evaluated via two cross-dataset tests, one involving all images from both DermNet and DermQuest datasets.

B. TUNING OF THE HYPERPARAMETER K

To determine an appropriate value for the segmentation hyperparameter K (i.e., the number of *dictionary atoms*), the proposed method was applied to all images in the DermIS

[2]This approach aligns with the experimental protocol established in [33] and [43].

TABLE 2. Summary of system parameters used in the proposed method.

Symbol	Description	Value(s)	Type
ω	Patch size (in pixels)	$\lfloor \frac{0.01L+2}{2} \rfloor - 1$	Adaptive
K	Number of dictionary atoms	3 (see Section III-B)	Tuned
σ_e	Width of the Gaussian kernel for computing graph weights	$\frac{1}{\sqrt{800}} \cdot \max_{j,j'} \|x_j^* - x_{j'}^*\|_2$ [42]	Adaptive
κ	Size of corner regions used in post-processing	$\lfloor \frac{1}{5} \cdot \max\{H, W\} \rfloor$ [33]	Adaptive
σ_1	Spread of Gaussian kernel for region-of-interest selection	$\lfloor \frac{1}{10} \cdot \max\{H, W\} \rfloor$ [33]	Adaptive
N/A	Kernel size and standard deviation for boundary smoothing	$3\omega \times 3\omega$; ω [33]	Adaptive
N/A	Global threshold for final mask generation	0.1 [33]	Fixed

dataset while varying K from 3 to 9 for each image. The method results obtained with $K \in \{3, 4, \ldots, 9\}$ were ranked from 1 (indicating the best performance) to 7 (indicating the worst performance) for each image according to the BF_1 score.

Based on the obtained rankings, the Friedman test was employed, resulting in a p-value of approximately 4.78%, and hence revealing significant performance differences between the proposed method variants for different K values. A Nemenyi post-hoc test was then performed, whose outcomes are graphically represented in the critical difference (CD) diagram shown in Fig. 6.

As depicted in Fig. 6, the proposed method tuned with K = 3 outperforms the alternative settings by achieving the lowest average ranking of 3.42. Conversely, when K is set to 9, the method underperforms the compared alternatives, attaining the highest average ranking of 4.66.

Remarkably, the Nemenyi post-hoc test found no significant differences between the proposed method with K = 3 and its alternatives with K values varying from 4 to 8. These alternatives reached average rankings of 3.80, 3.99, 4.05, 3.98, and 4.11, respectively. Similarly, no significant differences were identified among the method variations with K values varying from 4 to 9. These experimental results suggest that the proposed method exhibits limited sensitivity to the choice of K. Nevertheless, as K = 3 yielded the optimal observed average ranking, it is assumed that the proposed segmentation method utilizes K = 3 as the method hyperparameter.

C. ABLATION STUDY

To evaluate the specific contributions of each component for the proposed segmentation method, a comparative analysis was performed between the original proposal and its ablated versions. These versions, detailed below, were systematically crafted by removing or modifying a single building block within the diagram illustrated in Fig. 2 to identify their influence on the overall performance.

- Ablated version 1: This variant removes the first component (i.e., the preprocessing block);
- Ablated version 2: This variant modifies the second component (i.e., the saliency map estimation block) to simply obtain the intensity channel from the HSI color space;
- Ablated version 3: This variant modifies the third component (i.e., the dictionary learning block) to be initialized with the K-SVD method [44], with $K = 2\omega^2$ atoms (where ω denotes the patch size) and sparsity level $s_0 = 3$, instead of NMF;
- Ablated version 4: This variant modifies the fourth component (i.e., the clustering block) to use the K-means method, with K = 2 (initialized by the K-means++ algorithm) [45], instead of NGC;
- Ablated version 5: This variant removes the fifth component (i.e., the post-processing block).

Specifically, the proposed method and its ablated versions were applied to all images in the DermIS dataset, and each image segmentation result was ranked from 1 to 6 based on the BF_1 score. The obtained rankings underwent the Friedman test, revealing a p-value less than 5%, and indicating significant performance differences between the proposed method and its ablated versions. Subsequently, the Bonferroni-Dunn post-hoc test was conducted. The results of this post-hoc test are summarized in Fig. 7.

As can be observed in Fig. 7, the proposed segmentation method achieved the lowest average ranking (2.25), outperforming the ablated versions 1, 2, 5, 4, and 3, that have average rankings of 2.45, 3.22, 3.26, 4.36, and 5.46, respectively. The Bonferroni-Dunn post-hoc test confirms significant gain in performance of the proposed method over the compared ablated versions, except for ablated version 5. While not statistically superior to alternative 5, the proposed method exhibits a lower average ranking.

D. COMPARISON WITH THE STATE-OF-THE-ART

In order to assess comprehensively the cross-domain adaptability of the proposed segmentation method, it was compared with state-of-the-art approaches through two rigorous cross-dataset tests. One test utilized the entire image set from the DermNet dataset, while the other incorporated the complete set of images from DermQuest dataset. The following state-of-the-art representative segmentation alternatives were considered in the comparison:

- Dermoscopic Skin Network (DSNet) [29];
- Fully Convolutional Network (FCNet) [46];
- Full-resolution Convolutional Network (FrCNet) [30];
- Local/Global Patch-based Network (LGPNet) [31];
- Pyramid Scene Parsing Network (PSPNet) [21];

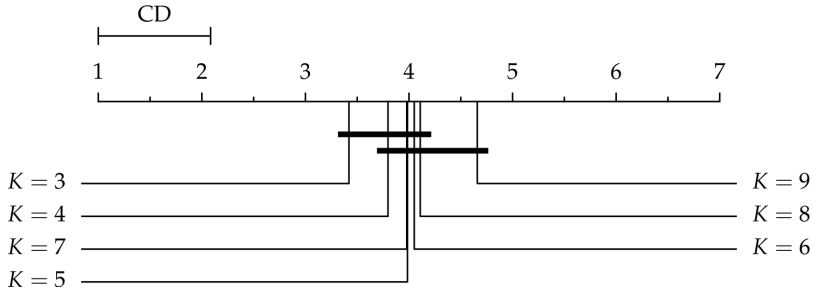


FIGURE 6. Average ranking comparison of different variants of the proposed segmentation method, with K varying from 3 to 9, against each other with the Nemenyi post-hoc test on the DermIS dataset and based on the BF_1 score (groups of variants that not significantly differ are connected by a thicker line).

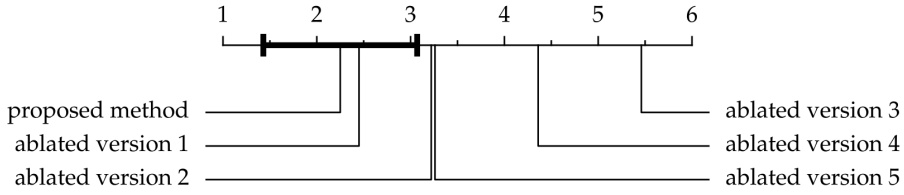
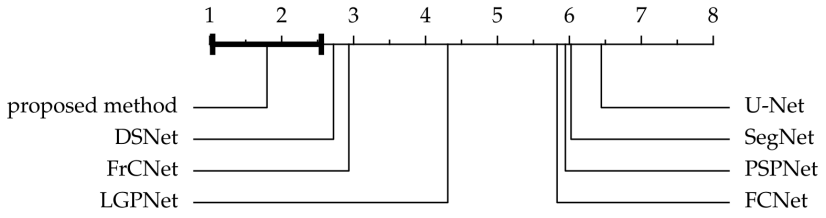
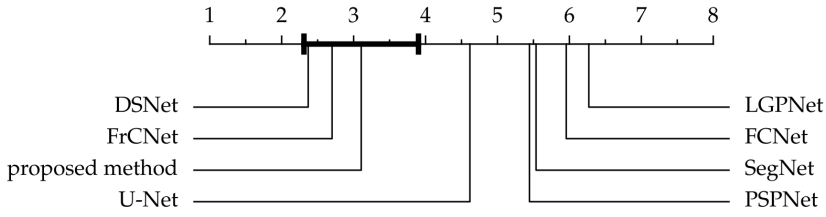


FIGURE 7. Average ranking comparison of the proposed segmentation method against its ablated versions with the Bonferroni-Dunn post-hoc test on the DermIS dataset and based on the BF_1 score (ablated versions with ranks outside the indicated interval significantly differ from the proposed method).



(a) On the DermNet dataset.



(b) On the DermQuest dataset.

FIGURE 8. Average ranking comparison of the proposed segmentation method against state-of-the-art representative segmentation alternatives with the Bonferroni-Dunn post-hoc test and based on the BF_1 score (alternatives with ranks outside the indicated interval significantly differ from the proposed method).

- Segmentation Network (SegNet) [22];
- U-shaped Network (U-Net) [20].

The evaluation process involved applying both the proposed method and the state-of-the-art alternatives to each image within the DermNet and DermQuest datasets. Subsequently, the resulting segmentations for each image were ranked from 1 to 8 based on the BF_1 score. The average rankings were then separately calculated as illustrated in Figs. 8a and 8b for the DermNet and DermQuest datasets, respectively.

The post-test results outlined in Fig. 8a indicate that, within the DermNet dataset, the proposed segmentation method significantly outperforms the compared alternatives. Conversely, the post-test findings in Fig. 8b suggest that, for the DermQuest dataset, the proposed method demonstrates comparable performance to DSNet and FrCNet, while statistically outperforming the other approaches under analysis.

IV. CONCLUSION

This work proposed a novel patch-based model for skin lesion segmentation in clinical images, which integrates nonnegative matrix factorization and mutual information to find an adequate dictionary that guides the segmentation process and improves the efficiency of the normalized graph cuts technique. When evaluated using the boundary F1 score measure, the experiments conducted suggest that the proposed method can statistically outperform both, traditional deep learning baselines and more recent alternatives specifically designed for skin lesion segmentation, across two distinct datasets (i.e., DermNet and DermQuest) that were used exclusively for cross-dataset testing.

While deep learning-based models typically rely on large labeled datasets and computationally intensive training with high-end graphics processing units, the proposed method is a fully unsupervised alternative that does not require pre-training, and offers inherent resilience to data biases. Moreover, deep learning-based approaches generally operate on image inputs of fixed dimensions, thereby requiring input images to be resized to use standard resolutions (e.g., 224 $\times$ 224 pixels). This process may lead to a loss of spatial detail and, consequently, to coarser segmentation results. In contrast, the proposed method operates directly on full-resolution images without downsampling or architectural constraints.

The aforementioned aspects not only suggest that the proposed method can be well-suited for integration into clinical skin lesion analysis systems, but also hint at its capacity to generalize across datasets with varying characteristics, thereby supporting its applicability in real-world clinical settings. However, these benefits come at the cost of relatively longer inference times for segmentation when compared to deep learning-based counterparts, particularly for very large input images. Further noteworthy limitations of the proposed method include the use of non-overlapping patches, which introduces block artifacts and requires additional post-processing, and the lack of evaluation on other dermatological modalities (e.g., dermoscopic images).

Future work shall focus on integrating Vision Transformers (ViTs) [47], [48], [49] with the proposed method. Specifically, by leveraging long-range contextual dependencies, the attention mechanism of ViTs may enable more informed and efficient patch clustering, thus accelerating inference and potentially improving segmentation accuracy. Another promising research direction involves investigating how the proposed method can be adapted for other medical imaging tasks, where its patch-based formulation and unsupervised nature could yield practical advantages.

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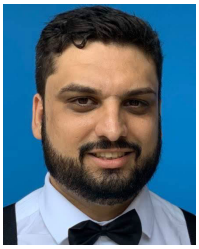
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